PROCURESPHERE 360

ENTITY RELATIONSHIP DIAGRAM

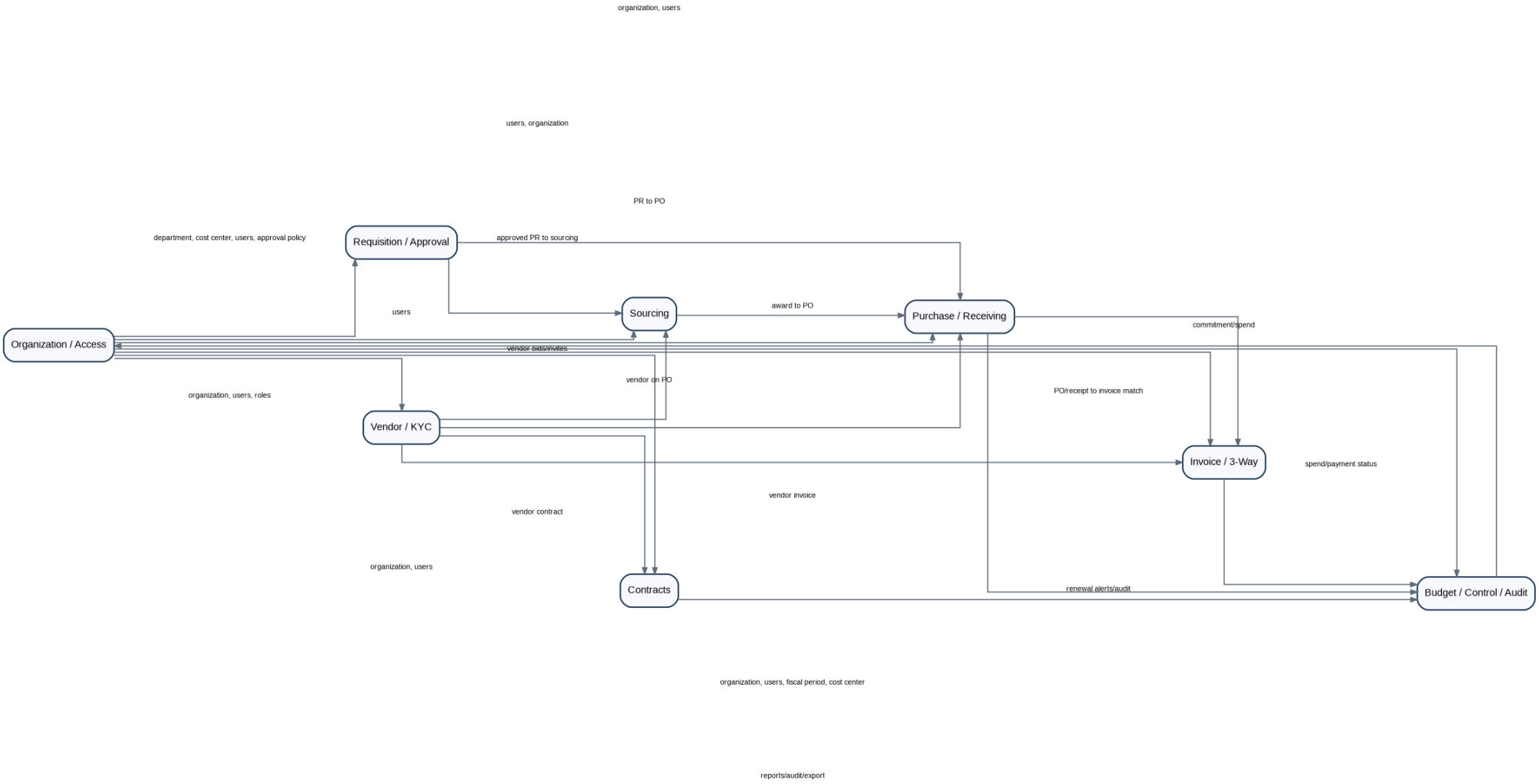
Normalized PostgreSQL Data Model | Version 1.0

A client-ready logical data model covering organization and access, vendor management, requisition and approvals, sourcing, purchasing and receiving, invoicing and 3-way matching, contracts, and budget/control/audit capabilities.

ERD Legend

PK	Primary key	**FK**	Foreign key	**UQ**	Unique constraint
IDX	Query index	**CHK**	Check constraint	**TS**	Timestamp

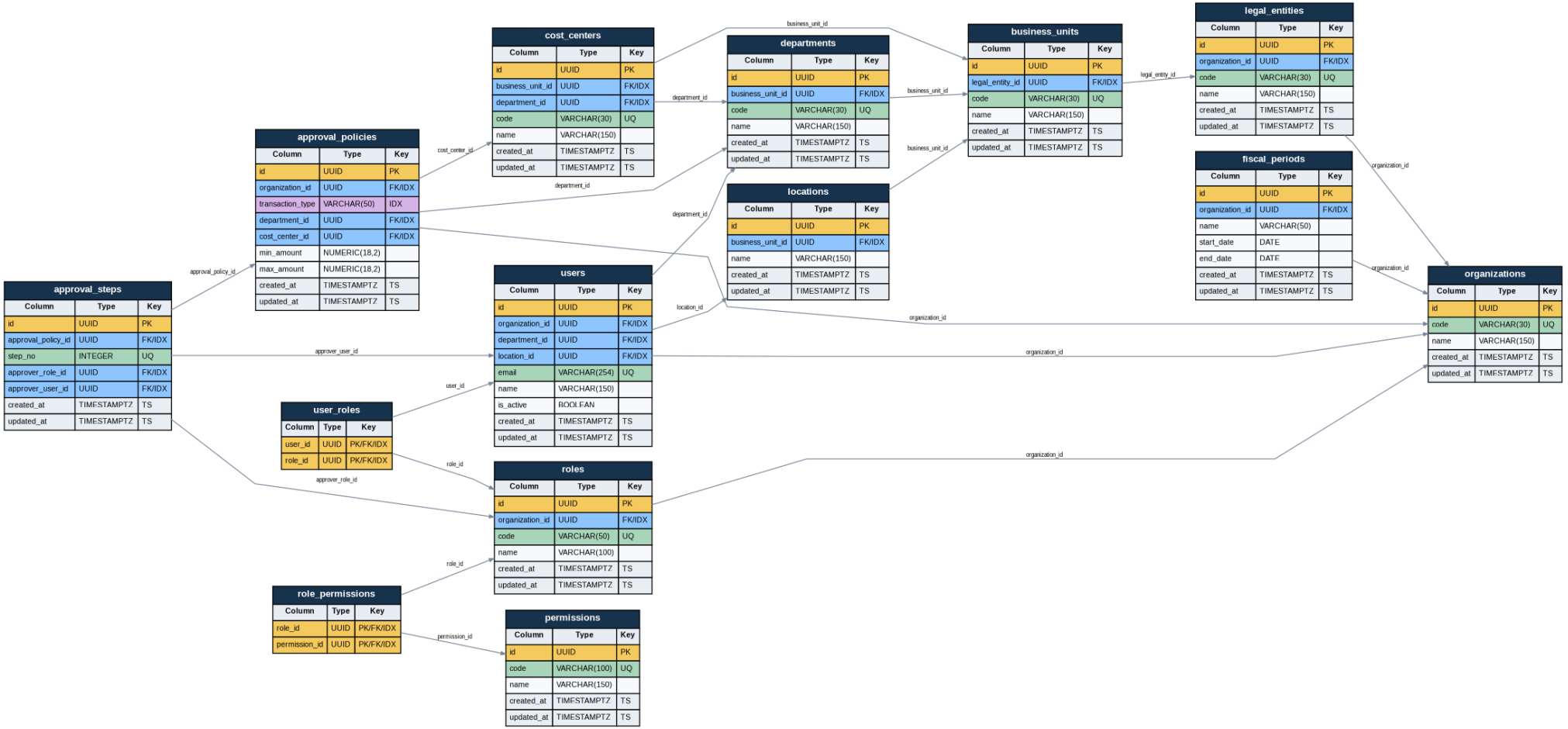
1. Master Domain ERD



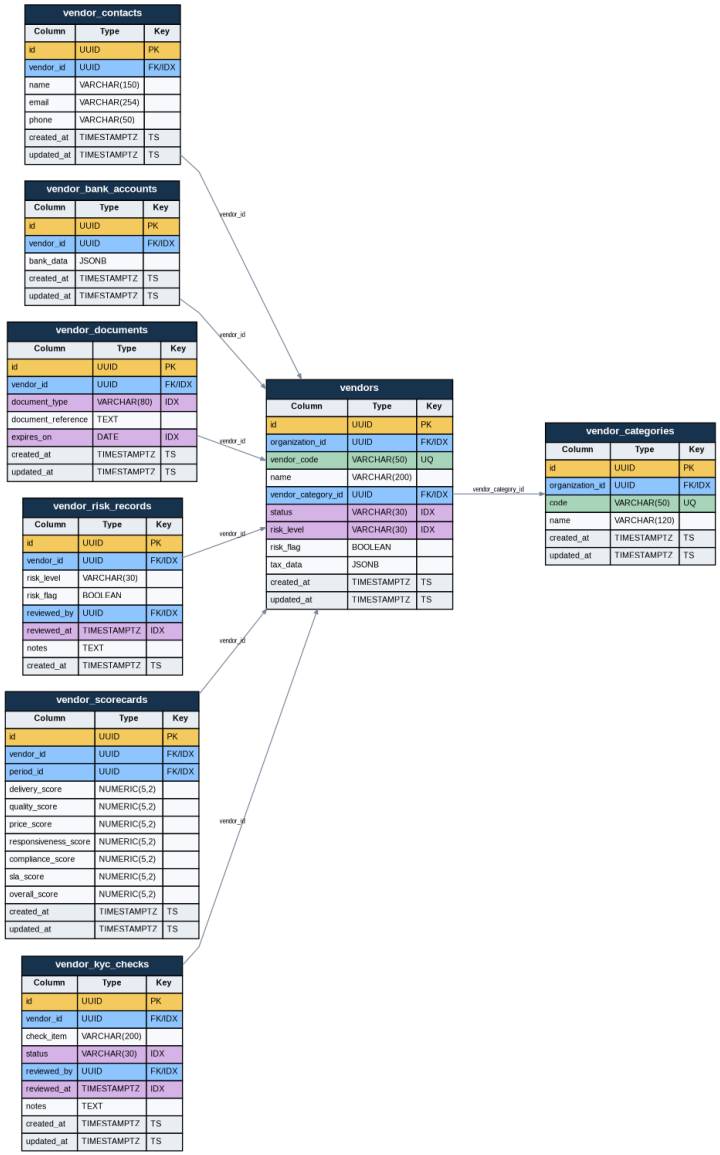
Domain-level view of the major business areas and their principal relationships.

2. Domain ERDs

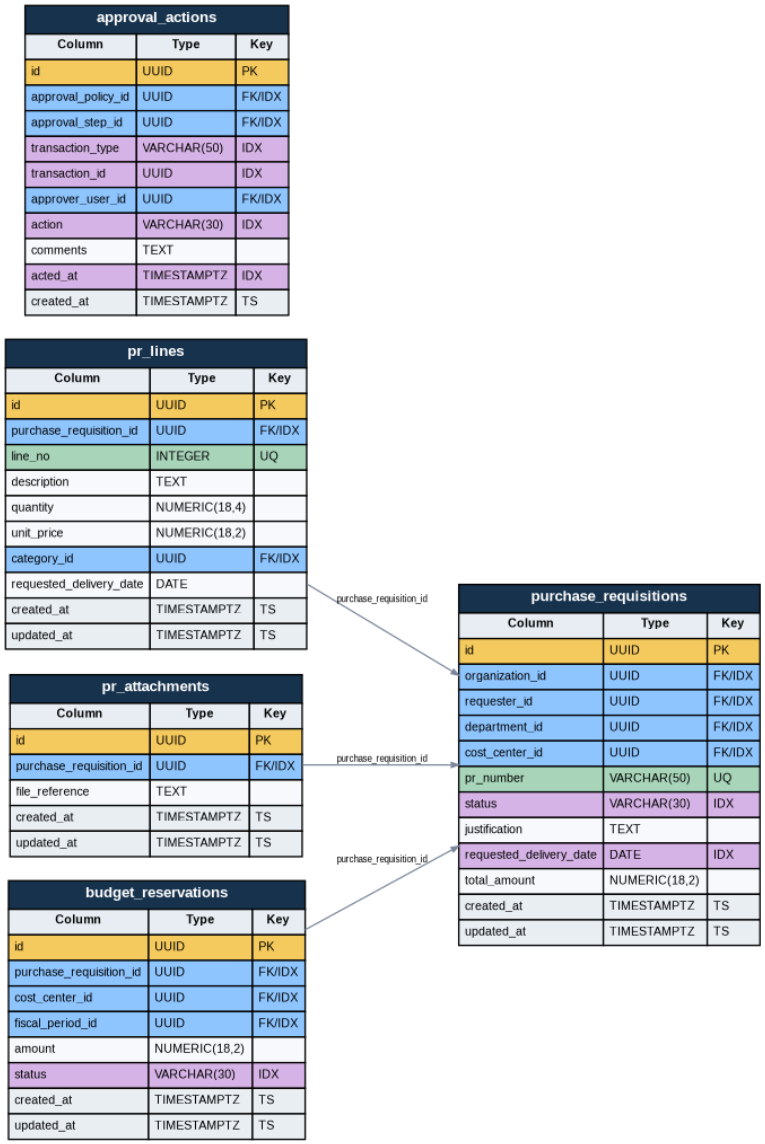
Organization / Access



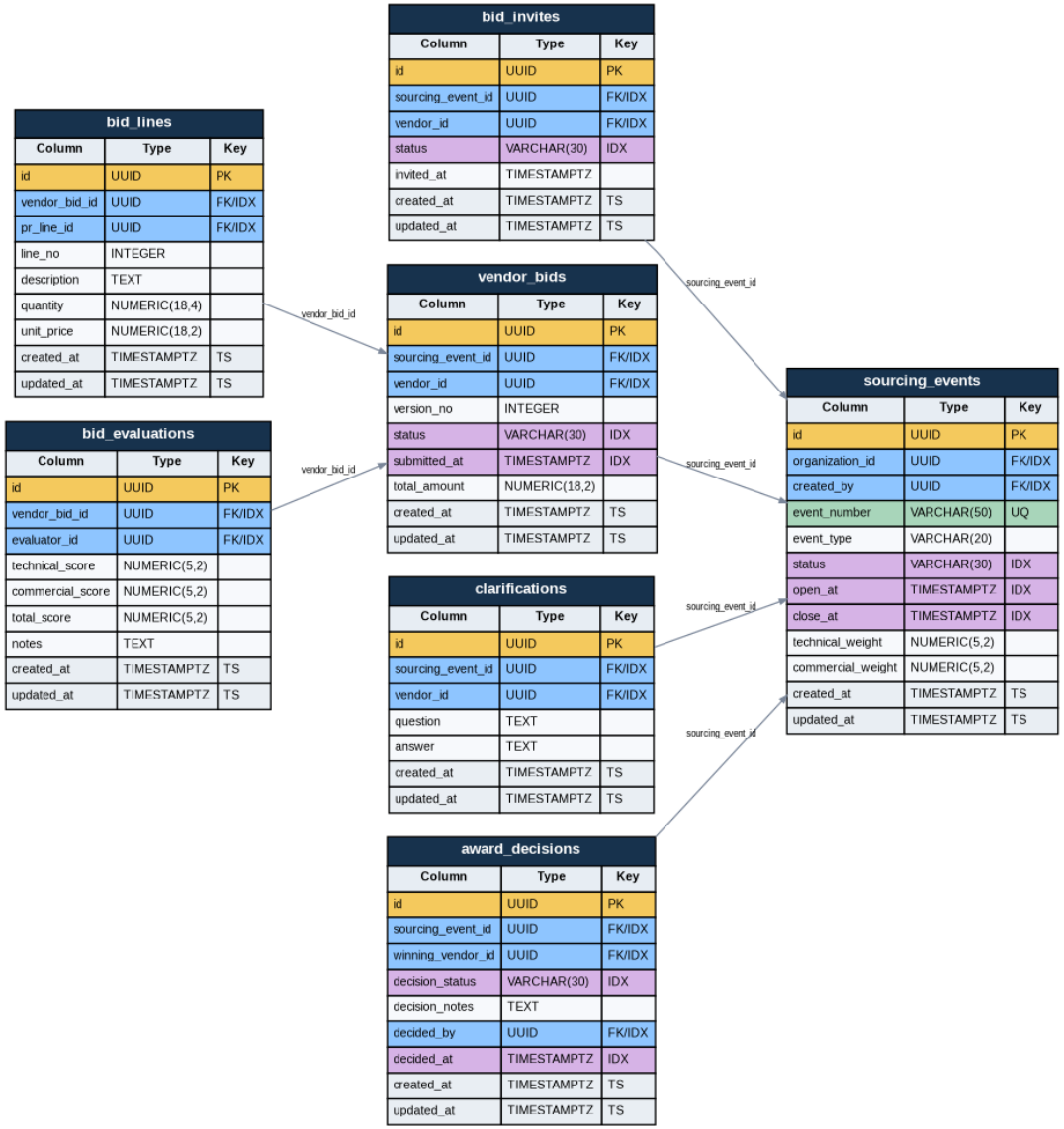
Vendor / KYC


vendor_contacts
Column Type Key
id UUID PK
vendor_id UUID FK/IDX
name VARCHAR(150)
email VARCHAR(254)
phone VARCHAR(50)
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendor_bank_accounts
Column Type Key
id UUID PK
vendor_id UUID FK/IDX
bank_data JSONB
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendor_documents
Column Type Key
id UUID PK
vendor_id UUID FK/IDX
document_type VARCHAR(80) IDX
document_reference TEXT
expires_on DATE IDX
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendor_risk_records
Column Type Key
id UUID PK
vendor_id UUID FK/IDX
risk_level VARCHAR(30)
risk_flag BOOLEAN
reviewed_by UUID FK/IDX
reviewed_at TIMESTAMPTZ IDX
notes TEXT
created_at TIMESTAMPTZ TS
vendor_scorecards
Column Type Key
id UUID PK
vendor_id UUID FK/IDX
period_id UUID FK/IDX
delivery_score NUMERIC(5,2)
quality_score NUMERIC(5,2)
price_score NUMERIC(5,2)
responsiveness_score NUMERIC(5,2)
compliance_score NUMERIC(5,2)
sla_score NUMERIC(5,2)
overall_score NUMERIC(5,2)
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendor_kyc_checks
Column Type Key
id UUID PK
vendor_id UUID FK/IDX
check_item VARCHAR(200)
status VARCHAR(30) IDX
reviewed_by UUID FK/IDX
reviewed_at TIMESTAMPTZ IDX
notes TEXT
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendors
Column Type Key
id UUID PK
organization_id UUID FK/IDX
vendor_code VARCHAR(50) UQ
name VARCHAR(200)
vendor_category_id UUID FK/IDX
status VARCHAR(30) IDX
risk_level VARCHAR(30) IDX
risk_flag BOOLEAN
tax_data JSONB
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendor_categories
Column Type Key
id UUID PK
organization_id UUID FK/IDX
code VARCHAR(50) UQ
name VARCHAR(120)
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendor_id
vendor_id
vendor_id
vendor_id
vendor_id
vendor_id
vendor_category_id

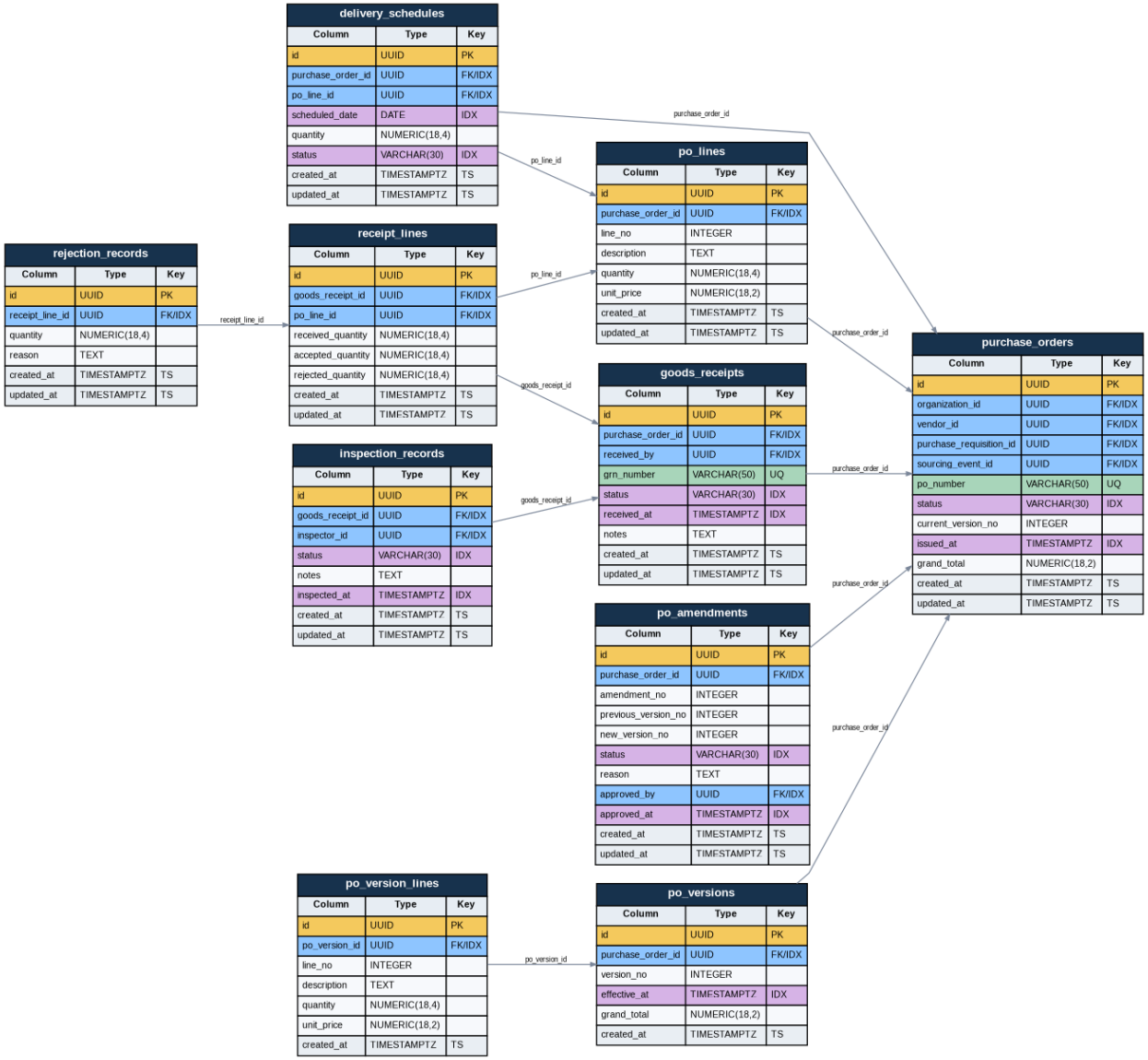
Requisition / Approval



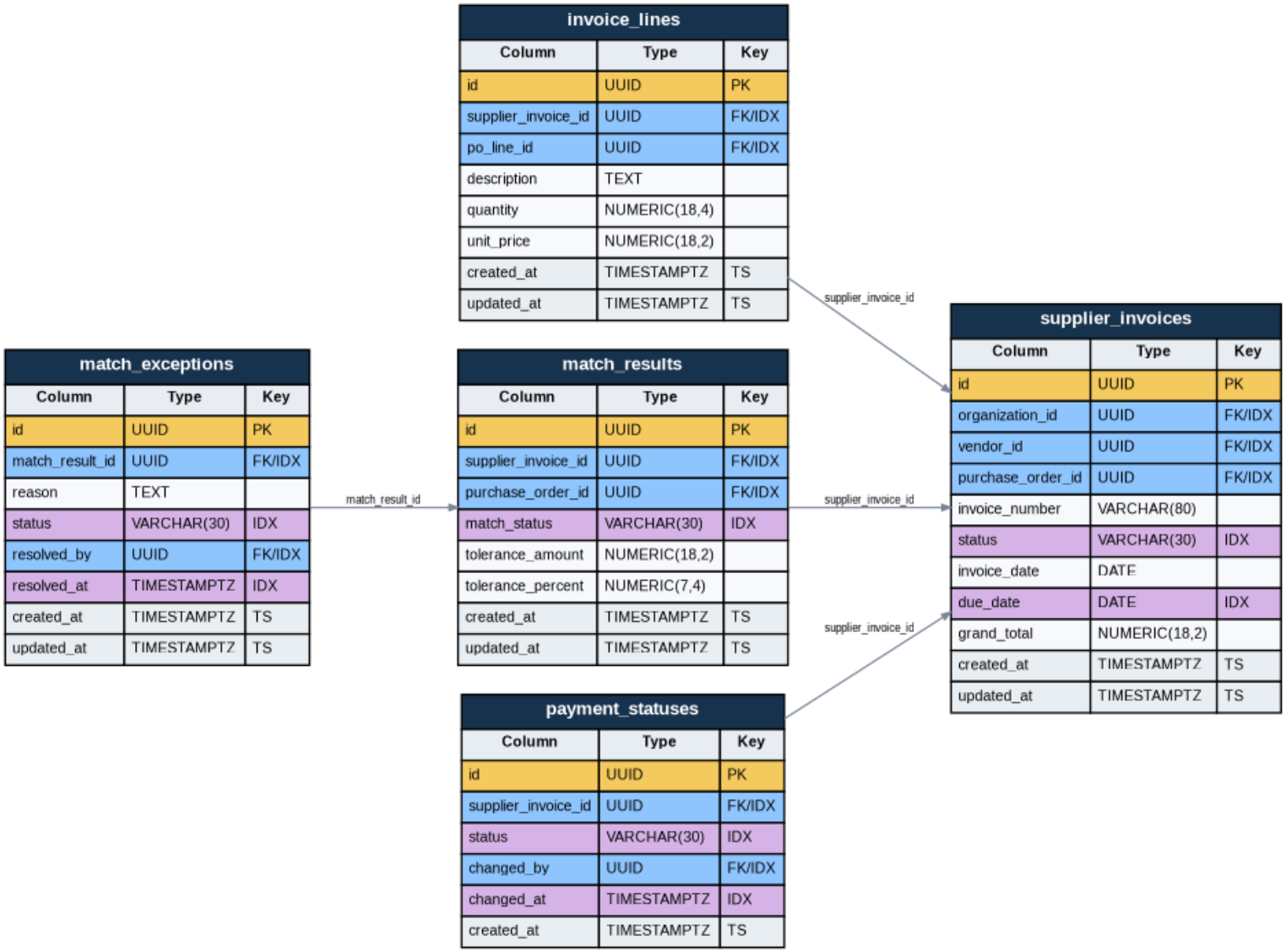
Sourcing


bid_invites
Column Type Key
id UUID PK
sourcing_event_id UUID FK/IDX
vendor_id UUID FK/IDX
status VARCHAR(30) IDX
invited_at TIMESTAMPTZ
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
bid_lines
Column Type Key
id UUID PK
vendor_bid_id UUID FK/IDX
pr_line_id UUID FK/IDX
line_no INTEGER
description TEXT
quantity NUMERIC(18,4)
unit_price NUMERIC(18,2)
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendor_bids
Column Type Key
id UUID PK
sourcing_event_id UUID FK/IDX
vendor_id UUID FK/IDX
version_no INTEGER
status VARCHAR(30) IDX
submitted_at TIMESTAMPTZ IDX
total_amount NUMERIC(18,2)
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
sourcing_events
Column Type Key
id UUID PK
organization_id UUID FK/IDX
created_by UUID FK/IDX
event_number VARCHAR(50) UQ
event_type VARCHAR(20)
status VARCHAR(30) IDX
open_at TIMESTAMPTZ IDX
close_at TIMESTAMPTZ IDX
technical_weight NUMERIC(5,2)
commercial_weight NUMERIC(5,2)
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
bid_evaluations
Column Type Key
id UUID PK
vendor_bid_id UUID FK/IDX
evaluator_id UUID FK/IDX
technical_score NUMERIC(5,2)
commercial_score NUMERIC(5,2)
total_score NUMERIC(5,2)
notes TEXT
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
clarifications
Column Type Key
id UUID PK
sourcing_event_id UUID FK/IDX
vendor_id UUID FK/IDX
question TEXT
answer TEXT
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
award_decisions
Column Type Key
id UUID PK
sourcing_event_id UUID FK/IDX
winning_vendor_id UUID FK/IDX
decision_status VARCHAR(30) IDX
decision_notes TEXT
decided_by UUID FK/IDX
decided_at TIMESTAMPTZ IDX
created_at TIMESTAMPTZ TS
updated_at TIMESTAMPTZ TS
vendor_bid_id
vendor_bid_id
sourcing_event_id
sourcing_event_id
sourcing_event_id
sourcing_event_id

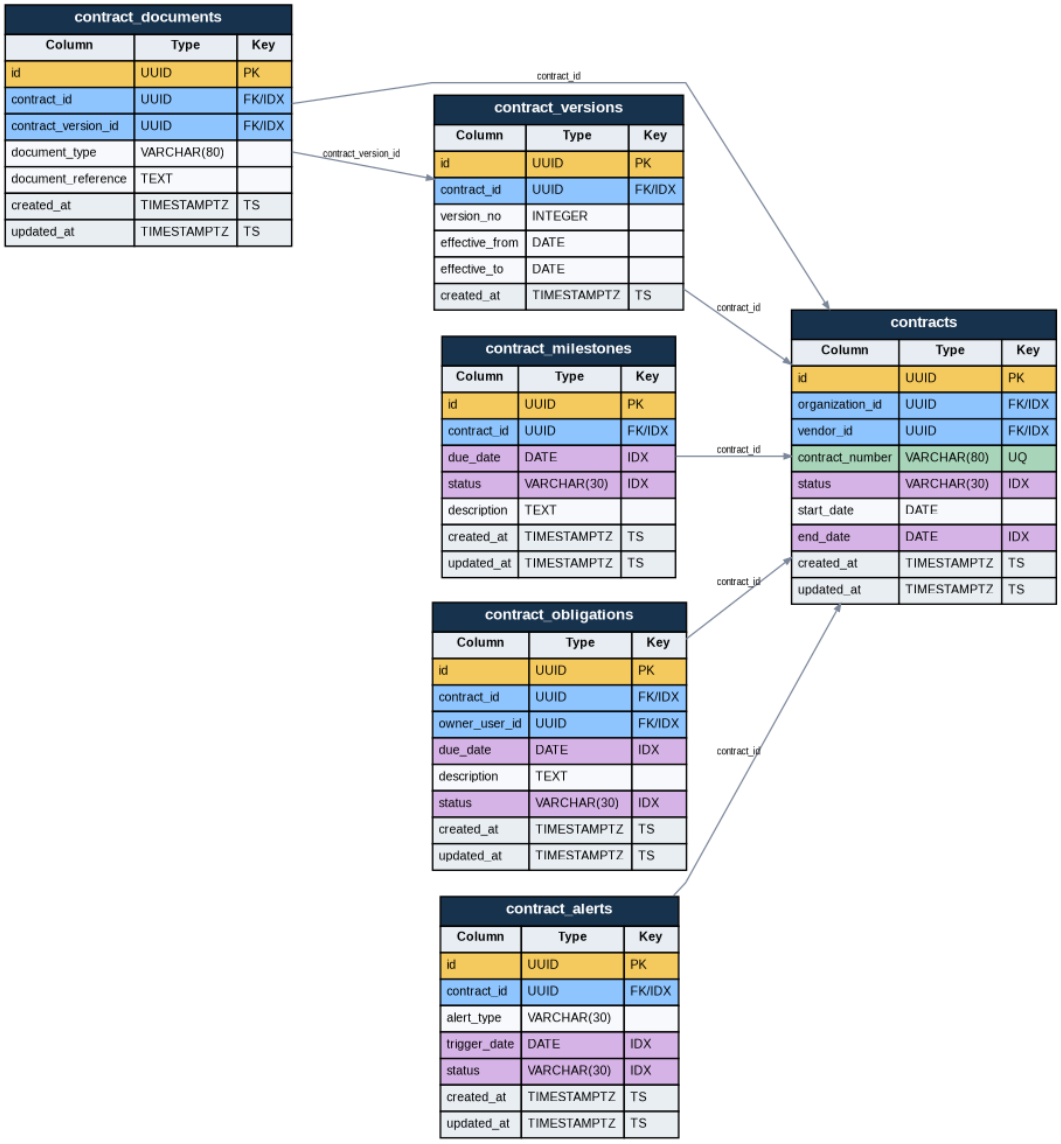
Purchase / Receiving



Invoice / 3-Way



Contracts


contract_documents
Column | Type | Key
id | UUID | PK
contract_id | UUID | FK/IDX
contract_version_id | UUID | FK/IDX
document_type | VARCHAR(80) |
document_reference | TEXT |
created_at | TIMESTAMPTZ | TS
updated_at | TIMESTAMPTZ | TS
contract_id
contract_version_id
contract_versions
Column | Type | Key
id | UUID | PK
contract_id | UUID | FK/IDX
version_no | INTEGER |
effective_from | DATE |
effective_to | DATE |
created_at | TIMESTAMPTZ | TS
contract_id
contract_milestones
Column | Type | Key
id | UUID | PK
contract_id | UUID | FK/IDX
due_date | DATE | IDX
status | VARCHAR(30) | IDX
description | TEXT |
created_at | TIMESTAMPTZ | TS
updated_at | TIMESTAMPTZ | TS
contract_id
contracts
Column | Type | Key
id | UUID | PK
organization_id | UUID | FK/IDX
vendor_id | UUID | FK/IDX
contract_number | VARCHAR(80) | UQ
status | VARCHAR(30) | IDX
start_date | DATE |
end_date | DATE | IDX
created_at | TIMESTAMPTZ | TS
updated_at | TIMESTAMPTZ | TS
contract_id
contract_obligations
Column | Type | Key
id | UUID | PK
contract_id | UUID | FK/IDX
owner_user_id | UUID | FK/IDX
due_date | DATE | IDX
description | TEXT |
status | VARCHAR(30) | IDX
created_at | TIMESTAMPTZ | TS
updated_at | TIMESTAMPTZ | TS
contract_id
contract_alerts
Column | Type | Key
id | UUID | PK
contract_id | UUID | FK/IDX
alert_type | VARCHAR(30) |
trigger_date | DATE | IDX
status | VARCHAR(30) | IDX
created_at | TIMESTAMPTZ | TS
updated_at | TIMESTAMPTZ | TS

Budget / Control / Audit

budgets		
Column	Type	Key
id	UUID	PK
organization_id	UUID	FK/IDX
cost_center_id	UUID	FK/IDX
fiscal_period_id	UUID	FK/IDX
category_id	UUID	FK/IDX
allocated_amount	NUMERIC(18,2)	
reserved_amount	NUMERIC(18,2)	
actual_amount	NUMERIC(18,2)	
status	VARCHAR(30)	IDX
created_at	TIMESTAMPTZ	TS
updated_at	TIMESTAMPTZ	TS

spend_ledger		
Column	Type	Key
id	UUID	PK
organization_id	UUID	FK/IDX
cost_center_id	UUID	FK/IDX
vendor_id	UUID	FK/IDX
category_id	UUID	FK/IDX
fiscal_period_id	UUID	FK/IDX
source_type	VARCHAR(30)	IDX
source_id	UUID	IDX
amount	NUMERIC(18,2)	
posted_at	TIMESTAMPTZ	IDX
created_at	TIMESTAMPTZ	TS

notifications		
Column	Type	Key
id	UUID	PK
recipient_user_id	UUID	FK/IDX
notification_type	VARCHAR(50)	
title	VARCHAR(200)	
message	TEXT	
source_type	VARCHAR(50)	IDX
source_id	UUID	IDX
status	VARCHAR(20)	IDX
sent_at	TIMESTAMPTZ	
read_at	TIMESTAMPTZ	
created_at	TIMESTAMPTZ	TS
updated_at	TIMESTAMPTZ	TS

audit_logs		
Column	Type	Key
id	UUID	PK
organization_id	UUID	FK/IDX
actor_user_id	UUID	FK/IDX
action	VARCHAR(50)	
object_type	VARCHAR(80)	IDX
object_id	UUID	IDX
previous_state	JSONB	
updated_state	JSONB	
source_ip	INET	
user_agent	TEXT	
created_at	TIMESTAMPTZ	IDX

export_jobs		
Column	Type	Key
id	UUID	PK
requested_by	UUID	FK/IDX
report_type	VARCHAR(50)	
filters	JSONB	
status	VARCHAR(20)	IDX
storage_reference	TEXT	
created_at	TIMESTAMPTZ	TS
updated_at	TIMESTAMPTZ	TS

3. Table Inventory

Concise reference to the entities represented in the diagrams and their principal integrity rules.

Organization / Access

Table	Purpose	Key constraints
organizations	Organization / Cost Center; representative entity: Organization	UNIQUE(code)
legal_entities	legal entities	UNIQUE(organization_id, code)
business_units	units	UNIQUE(legal_entity_id, code)
departments	departments	UNIQUE(business_unit_id, code)
locations	locations	FK/IDX as applicable
cost_centers	cost centers	UNIQUE(business_unit_id, code)
fiscal_periods	fiscal periods	UNIQUE(organization_id, name); CHECK(start_date <= end_date)
users	User; primary users/roles in section 3.2	UNIQUE(email)
roles	Role; roles in section 3.2	UNIQUE(organization_id, code)
permissions	Permission; backend authorization requirement	UNIQUE(code)
user_roles	users ↔ roles RBAC relationship	PRIMARY KEY(user_id, role_id)
role_permissions	roles ↔ permissions RBAC relationship	PRIMARY KEY(role_id, permission_id)
approval_policies	ApprovalPolicy; rules by amount, department, cost center, transaction type	CHECK(min_amount IS NULL OR min_amount >= 0); CHECK(max_amount IS NULL OR max_amount >= 0); CHECK(max_amount IS NULL OR min_amount IS NULL OR max_amount >= min_amount)
approval_steps	ApprovalStep; ordered policy levels	UNIQUE(approval_policy_id, step_no); CHECK(step_no > 0); CHECK(approver_role_id IS NOT NULL OR approver_user_id IS NOT NULL)

Vendor / KYC

Table	Purpose	Key constraints
vendors	Vendor; onboarding/KYC/master/risk requirements	UNIQUE(organization_id, vendor_code)
vendor_contacts	VendorContact; vendor contact data	FK/IDX as applicable
vendor_bank_accounts	vendor bank data; separate table only to keep repeating bank data normalized	FK/IDX as applicable
vendor_documents	VendorDocument; KYC documents/supporting evidence	FK/IDX as applicable
vendor_categories	VendorCategory; vendor categories	UNIQUE(organization_id, code)
vendor_risk_records	VendorRiskRecord; due diligence/risk history	FK/IDX as applicable
vendor_scorecards	VendorScorecard; supplier performance history	CHECK(overall_score IS NULL OR overall_score BETWEEN 0 AND 100)

Table	Purpose	Key constraints
vendor_kyc_checks	due-diligence checklist; per-vendor checklist results	FK/IDX as applicable

Requisition / Approval

Table	Purpose	Key constraints
purchase_requisitions	PurchaseRequisition; PR workflow	UNIQUE(organization_id, pr_number); CHECK(total_amount >= 0)
pr_lines	PRLine; line items/services	UNIQUE(purchase_requisition_id, line_no); CHECK(quantity > 0); CHECK(unit_price >= 0)
pr_attachments	PRAttachment	FK/IDX as applicable
approval_actions	ApprovalAction; approval decisions/state trace	FK/IDX as applicable
budget_reservations	BudgetReservation; PR budget reservation	CHECK(amount > 0)

Sourcing

Table	Purpose	Key constraints
sourcing_events	SourcingEvent; RFQ/RFP lifecycle	UNIQUE(organization_id, event_number); CHECK(open_at IS NULL OR close_at IS NULL OR open_at < close_at)
bid_invites	BidInvite; invite eligible vendors	UNIQUE(sourcing_event_id, vendor_id)
vendor_bids	VendorBid; sealed bid response	UNIQUE(sourcing_event_id, vendor_id, version_no); CHECK(version_no > 0); CHECK(total_amount >= 0)
bid_lines	BidLine; commercial bid details	UNIQUE(vendor_bid_id, line_no); CHECK(quantity > 0); CHECK(unit_price >= 0)
bid_evaluations	BidEvaluation; weighted technical/commercial evaluation	UNIQUE(vendor_bid_id, evaluator_id); CHECK(total_score IS NULL OR total_score BETWEEN 0 AND 100)
clarifications	Clarification; sourcing clarification questions	FK/IDX as applicable
award_decisions	AwardDecision; award approval/decision audit	UNIQUE(sourcing_event_id)

Purchase / Receiving

Table	Purpose	Key constraints
purchase_orders	PurchaseOrder; PO workflow and versioning anchor	UNIQUE(organization_id, po_number); CHECK(current_version_no > 0); CHECK(grand_total >= 0)
po_lines	POLine; current PO lines	UNIQUE(purchase_order_id, line_no); CHECK(quantity > 0); CHECK(unit_price >= 0)
po_amendments	POAmendment; amendment/change-order control	UNIQUE(purchase_order_id, amendment_no); CHECK(new_version_no = previous_version_no + 1)
delivery_schedules	DeliverySchedule; planned delivery schedule	CHECK(quantity > 0)

Table	Purpose	Key constraints
po_versions	PO versioning; preserves prior versions	UNIQUE(purchase_order_id, version_no); CHECK(version_no > 0); CHECK(grand_total >= 0)
po_version_lines	preserve prior PO versions; normalized version-line snapshot	UNIQUE(po_version_id, line_no); CHECK(quantity > 0); CHECK(unit_price >= 0)
goods_receipts	GoodsReceipt; partial/final receipt	UNIQUE(grn_number)
receipt_lines	ReceiptLine; received/accepted/rejected quantities	CHECK(received_quantity >= 0); CHECK(accepted_quantity >= 0); CHECK(rejected_quantity >= 0); CHECK(accepted_quantity + rejected_quantity <= received_quantity)
inspection_records	InspectionRecord; quality acceptance/rejection	FK/IDX as applicable
rejection_records	RejectionRecord; rejected receipt quantities/evidence	CHECK(quantity > 0)

Invoice / 3-Way

Table	Purpose	Key constraints
supplier_invoices	SupplierInvoice; invoice capture and lifecycle	UNIQUE(vendor_id, invoice_number); CHECK(grand_total >= 0)
invoice_lines	InvoiceLine; invoice quantities/amounts linked to PO	CHECK(quantity > 0); CHECK(unit_price >= 0)
match_results	MatchResult; PO + receipt + invoice comparison	UNIQUE(supplier_invoice_id)
match_exceptions	MatchException; discrepancy workflow	FK/IDX as applicable
payment_statuses	PaymentStatus; payment lifecycle history	FK/IDX as applicable

Contracts

Table	Purpose	Key constraints
contracts	Contract; contract lifecycle	UNIQUE(organization_id, contract_number); CHECK(start_date <= end_date)
contract_versions	ContractVersion; versioning	UNIQUE(contract_id, version_no); CHECK(effective_to IS NULL OR effective_from <= effective_to)
contract_milestones	ContractMilestone; milestones	FK/IDX as applicable
contract_obligations	ContractObligation; obligations/SLA controls	FK/IDX as applicable
contract_alerts	ContractAlert; expiry/renewal alerts	UNIQUE(contract_id, alert_type, trigger_date)
contract_documents	ContractDocument; document vault	FK/IDX as applicable

Budget / Control / Audit

Table	Purpose	Key constraints
budgets	Budget; allocation/commitment/actual spend	CHECK(allocated_amount >= 0); CHECK(reserved_amount >= 0); CHECK(actual_amount >= 0)

Table	Purpose	Key constraints
spend_ledger	SpendLedger; actual spend analytics/control	CHECK(amount >= 0)
notifications	Notification; in-app/email alerts and escalation	FK/IDX as applicable
audit_logs	AuditLog; actor/action/object/state/source/timestamp	FK/IDX as applicable
export_jobs	ExportJob; report/audit export	FK/IDX as applicable

4. Design Notes & Items for Confirmation

• Cardinalities beyond the relationships needed for the named workflows are not specified and should be confirmed where business rules require additional relationship detail.

• Vendor tax, bank, and contact information is logically grouped in the model without introducing separate tax or banking master schemas where the underlying business structure has not been defined.

• Approval routing supports configurable approvers and workflow steps. A separate delegation data model is not included; delegation and approver-unavailability handling should be confirmed during detailed workflow design.

• Configurable 3-way-match tolerances are represented on match results. A separate tolerance master is not introduced.

• Reporting and export requirements are supported through the operational model, audit data, and export jobs. A separate reporting warehouse or star-schema layer is not included in this logical ERD.

5. Modeling Conventions

UUID identifiers are used for entity keys. Mutable business entities carry created_at and updated_at timestamps. Junction tables support normalized many-to-many relationships where required. Historical/versioned structures preserve prior states rather than overwriting transactional history. Polymorphic audit, notification, and ledger source references are intentionally represented as type-plus-identifier pairs where a single relational foreign key cannot span heterogeneous business objects.